

Biomedical Computer Vision

Image Segmentation

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Syllabus:

17.03

- Data dimensionality, image formats
- Image preprocessing

19.03

- Introduction to convolution
- Padding, stride, pooling

24.03

- Convolution-based operation: filtering (denoising, sharpening, edge detection)

27.03

- Introduction to segmentation: thresholding, masking, k-means...
- Post-processing: connected components, erosion, dilation...

31.03

- Introduction to Deep Learning
- Fully Connected Neural Networks, Convolutional Neural Networks

02.04

- NN training and regularization (dropout, weight decay...)

14.04

- Image segmentation with Deep Learning
- UNet

16.04

- CNN and others for other tasks (generation, denoising...)
- Review of the state of the art

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Image segmentation

Image segmentation is the task of clustering parts of an image together that belong to the same object class. This process is also called pixel-level classification, since it consists in assigning a label to each pixel in the image.

Segmentation usually acts as a pre-processing step to precisely delineate shape, size and boundaries of tumors and organs. This is useful to precisely quantify tumor volume, appropriately plan the treatment, and monitor how tumor evolves during therapy.

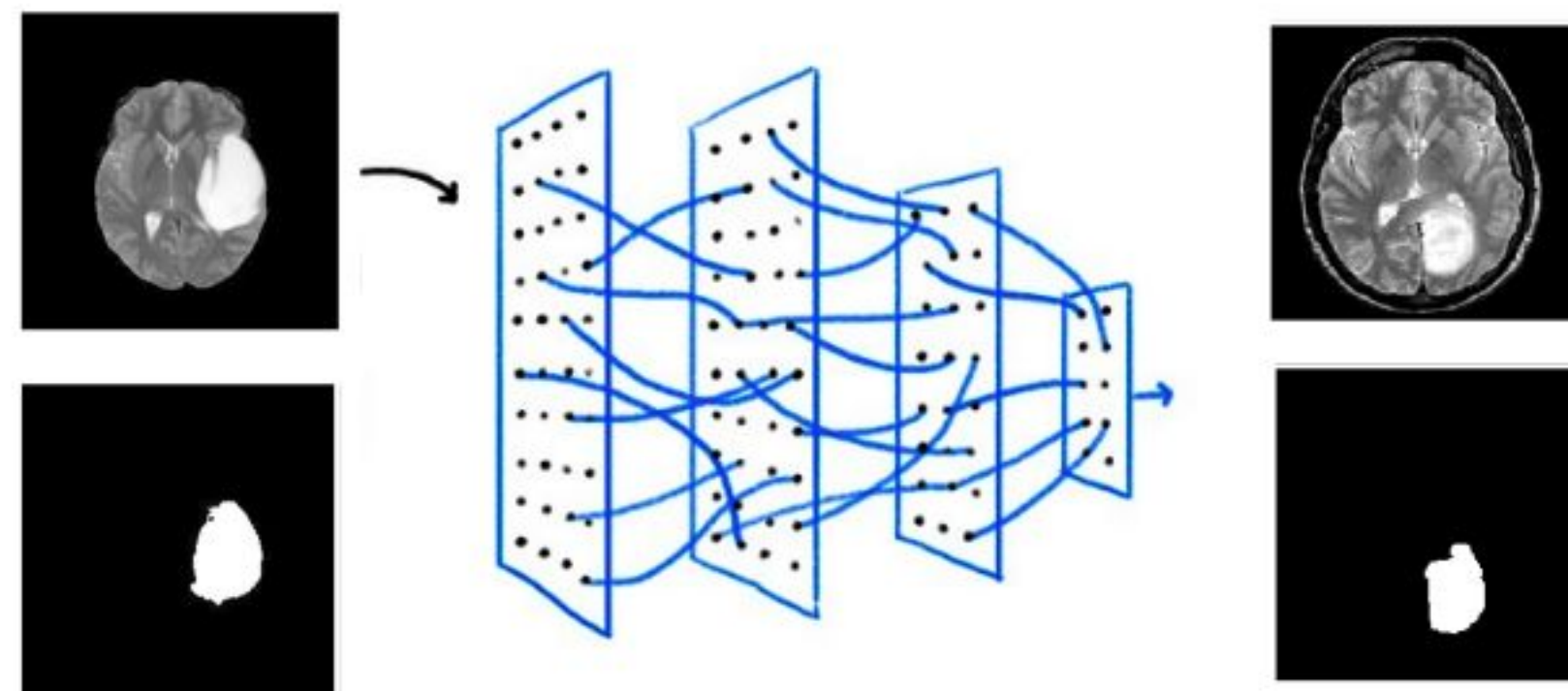


Image segmentation

Before Deep Learning became the standard, segmentation relied on semi-automatic methods such as Otsu thresholding, or region growing. These techniques are fast and simple, but they work through fixed rules based mainly on intensity or edges. As soon as images become noisy, low-contrast, or highly variable across scanners and patients, these rules fail. They have no understanding of texture, shape, or anatomical context, so they struggle with complex structures and fuzzy boundaries.

With Deep Learning, neural networks learn patterns directly from data: how tissues look, how lesions vary, and how global context relates to local details. This makes them far more robust to variability and much better at capturing subtle, clinically relevant boundaries.

Types of image segmentation

Image segmentation can be:

1. **Semantic segmentation:** associates every pixel of an image with a class label, but we don't distinguish between different objects of the same class. For example, if there are two cats in the image, both will be labeled as "cat" without any distinction between them.
2. **Instance segmentation:** treats multiple objects of the same class as distinct individual instances. So, if there are two cats in the image, instance segmentation will treat them as two distinct objects, even though they belong to the same class; it is closely related to object detection. In this case, we don't label all pixels in the image, we are only interested in boundaries of specific objects.
3. **Panoptic segmentation:** combines semantic and instance segmentation, by assigning a class label to each pixel, and by segmenting each object instance. Every pixel is assigned a class, but if there are multiple instances of a class, we know which pixel belong to which instance of the class.

Segmentation architectures

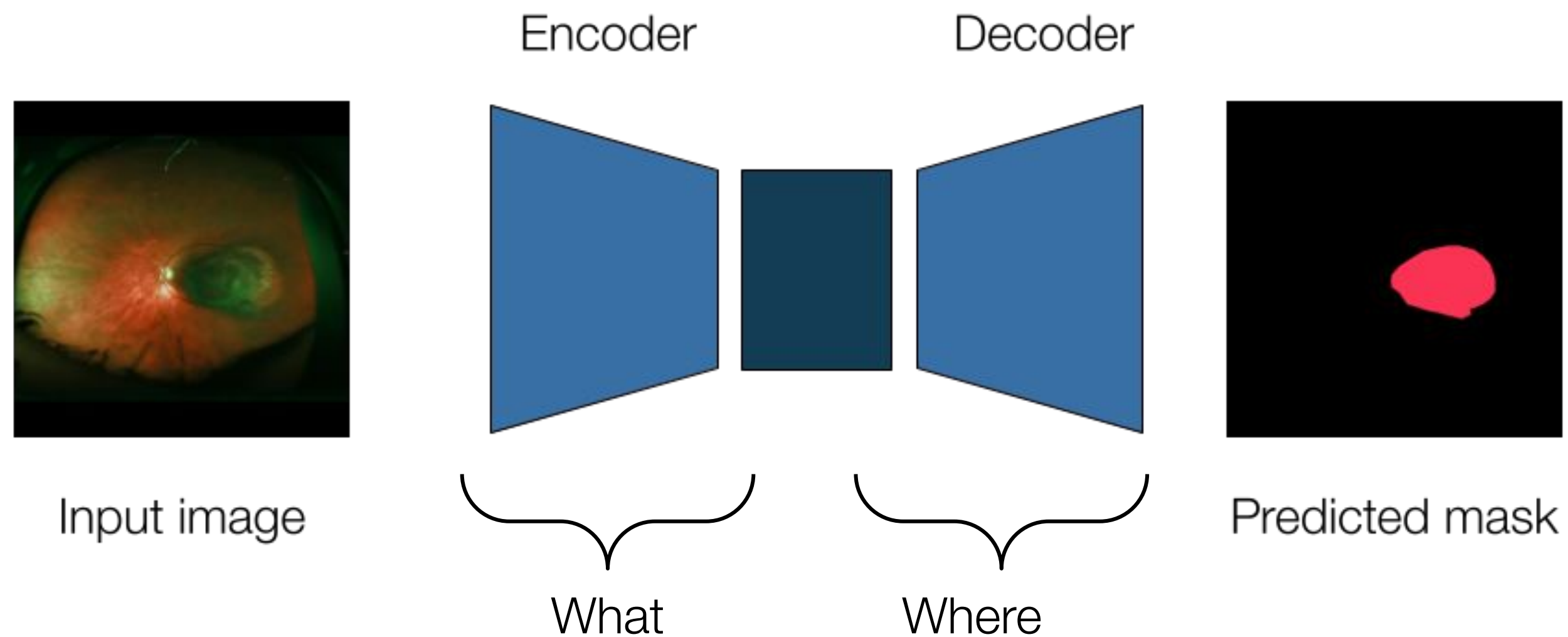
When talking about segmentation, there is an inherent tension between semantics and location:

- On the one hand we need to go deep, to extract high-level semantic information from the image.
- On the other hand we want to stay local to not lose spatial resolution and know where things are.

Therefore, we don't use the same architectures employed for the classification task, but we adopt an **encoder-decoder** architecture. The first part is the same of a standard classification network; the second half is meant to upsample the predictions to cover each pixel in the image.

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Segmentation architectures

Segmentation is a dense-prediction task. For this reason, all the image classification neural network architectures can be exploited, taking care of substituting the last fully connected layer with 1×1 convolutional layers with transposed convolutions for upsampling, linear activation functions and a SoftMax layer to regain spatial context.

- A 1×1 convolution acts like a fully connected layer applied at every pixel, converting deep feature vectors into class scores without destroying spatial layout. It preserves the $H \times W$ map while reducing channels to C (number of classes), enabling the decoder to upsample these coarse class logits back to full resolution.
- After the 1×1 convolution, we obtain a very low-resolution “segmentation map”. Transposed convolutions (or learned upsampling layers) progressively recover spatial resolution, coming back to the full-image size

U-Net

U-Net: Convolutional Networks for Biomedical Image Segmentation

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U-Net

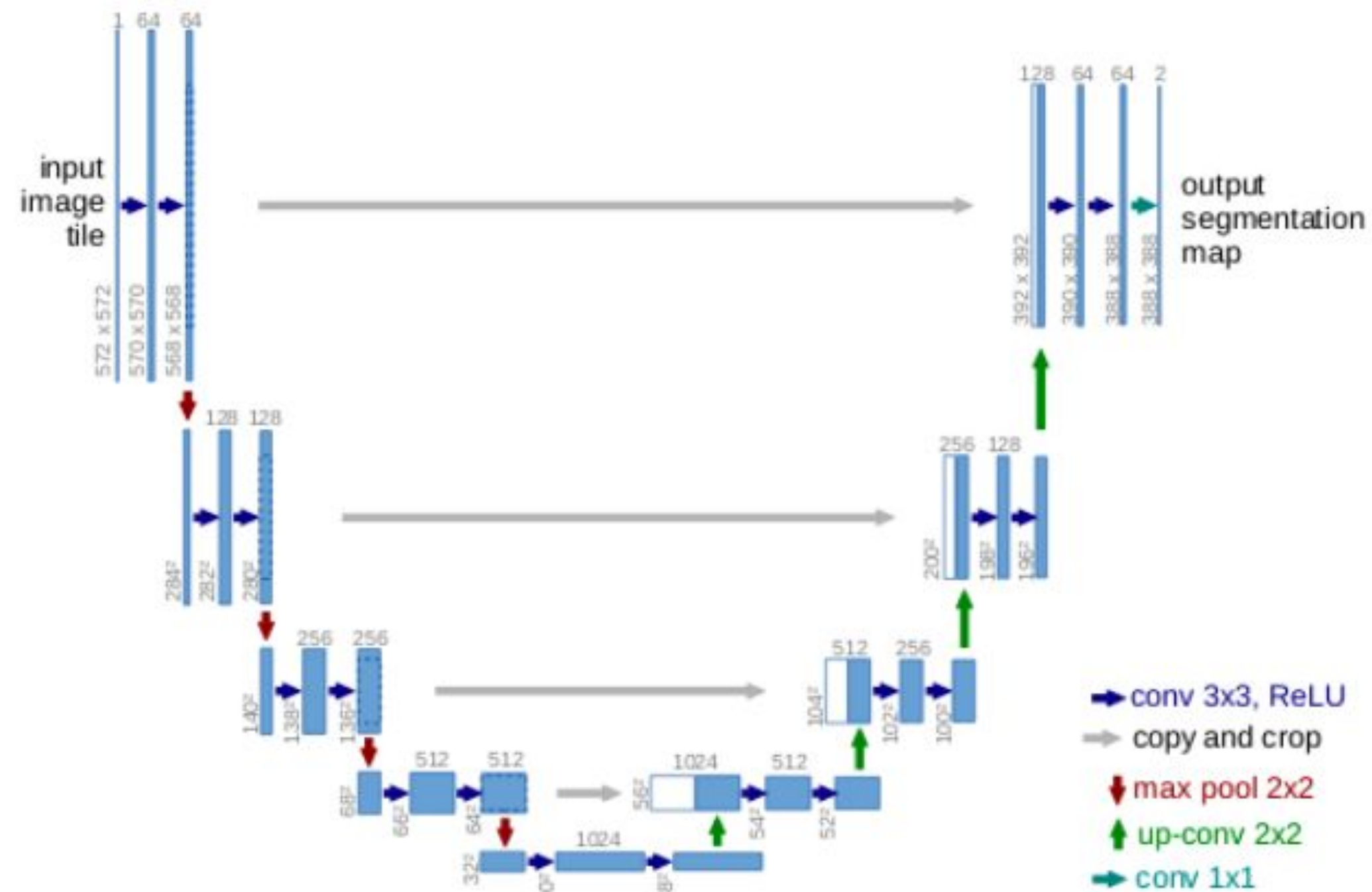


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

U-Net

U-Net is a convolutional neural network architecture developed for image segmentation, specifically designed for biomedical image segmentation tasks.

U-Net builds upon the concept of Fully Convolutional Neural Networks (FCN).

Its name comes from its U-shaped design. It follows the concept of encoder-decoder previously described, and it is composed of three parts.

- Encoder: the contracting path, which captures context through repeated convolutions and max pooling.
- Horizontal bottleneck: serves as the bridge between the encoder (context extraction) and decoder (spatial reconstruction).
- Decoder: the expanding or up-sampling path, complementary to the encoder, which enables the restoration of spatial dimensions.

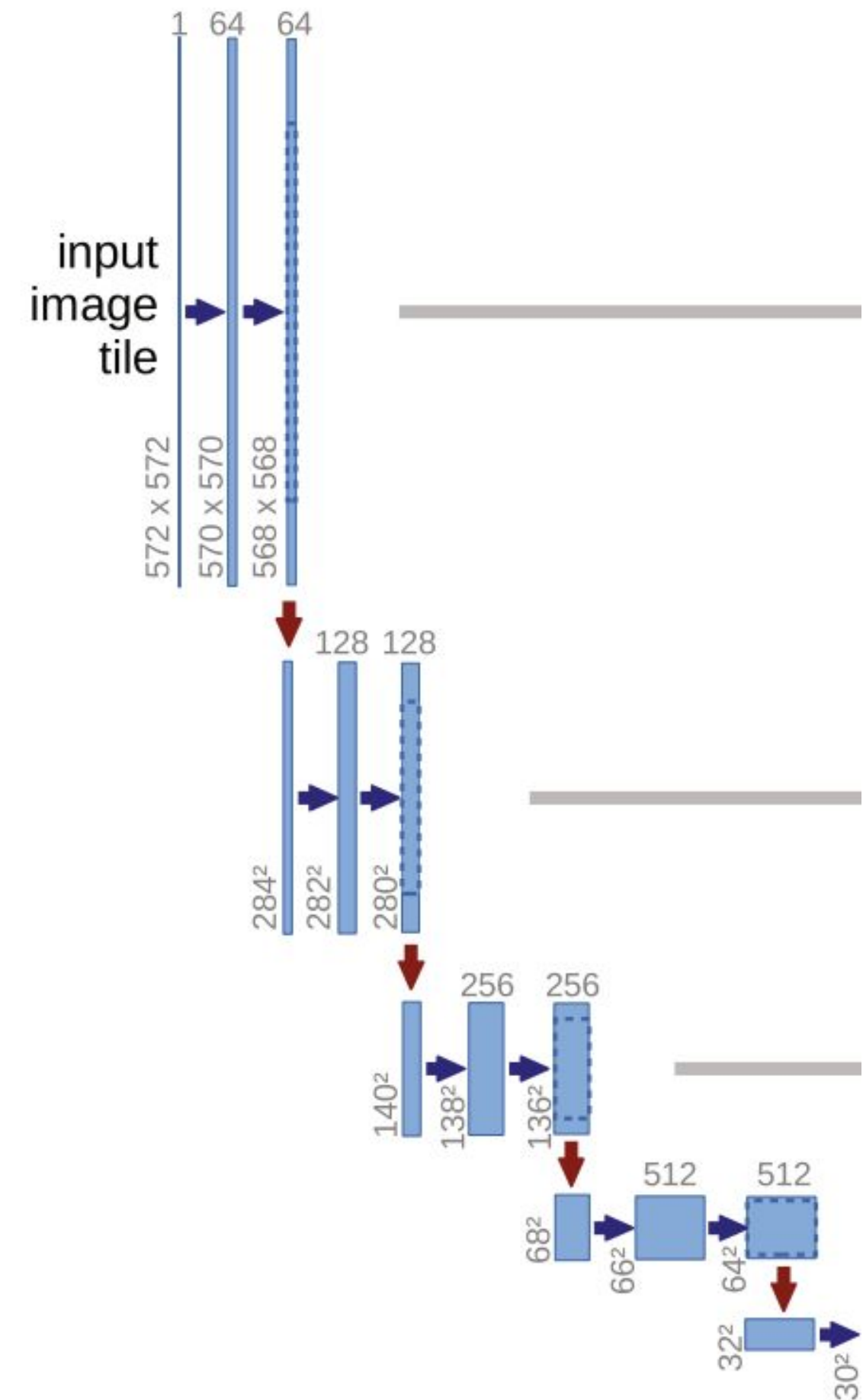
U-Net also has skip connections in order to improve localization. The upsampled output is concatenated with the corresponding cropped (cropped due to the loss of border pixels in every convolution) feature maps from the contracting path (the features learned during downsampling are used during upsampling).

U-Net: Contracting path

Repeats blocks of:

- 3x3 convolutions + ReLU
- 3x3 convolution + ReLU
- 2x2 max pooling, with stride 2 for downsampling

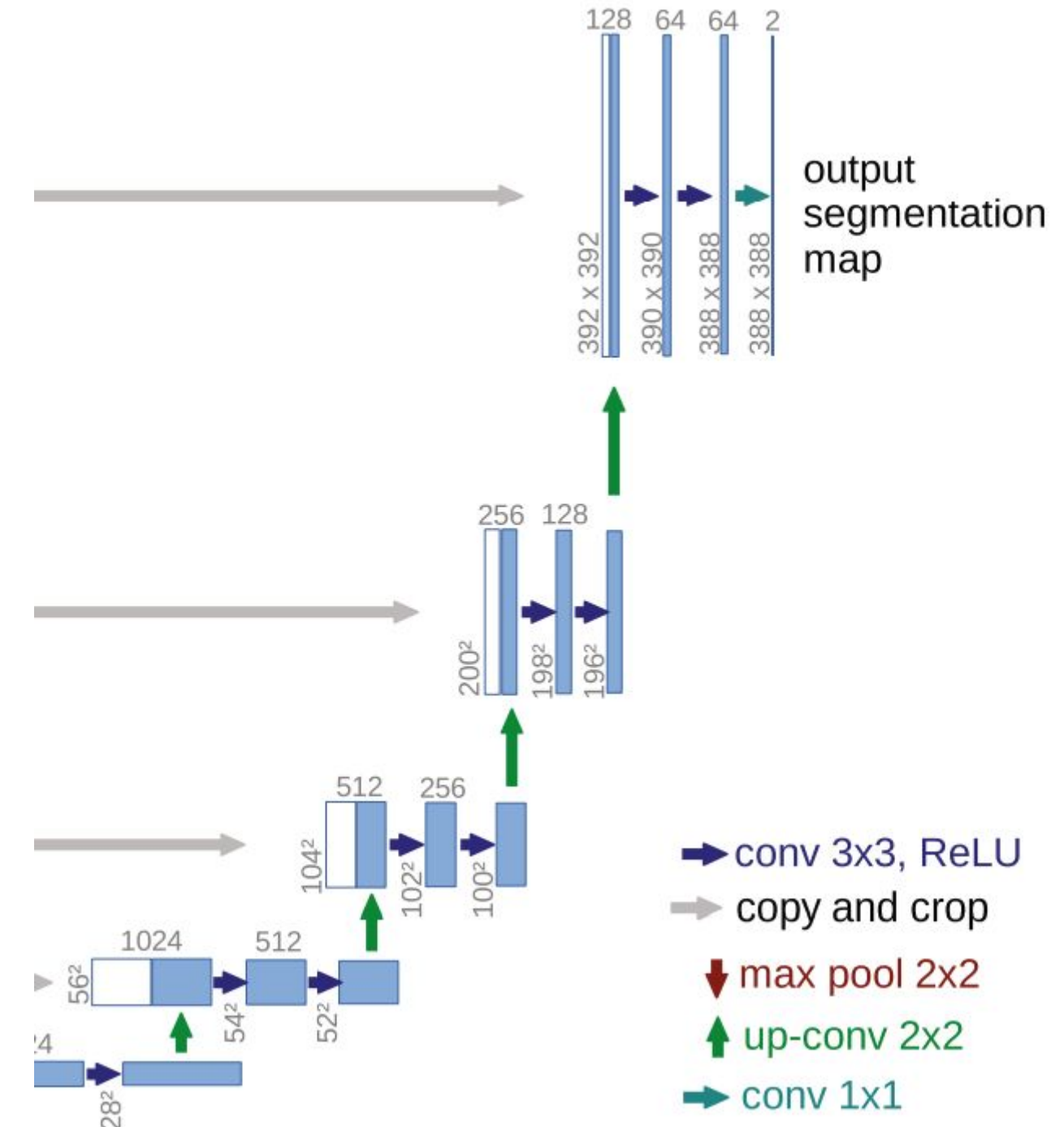
At each downsampling step, the number of feature maps is doubled.



U-Net: Expanding path

Repeats blocks of:

- 2x2 transposed convolutions, halving the number of feature maps, but increasing the spatial dimension
- Concatenation of corresponding cropped features from the encoder.
- 3x3 convolution + ReLU
- 3x3 convolution + ReLU

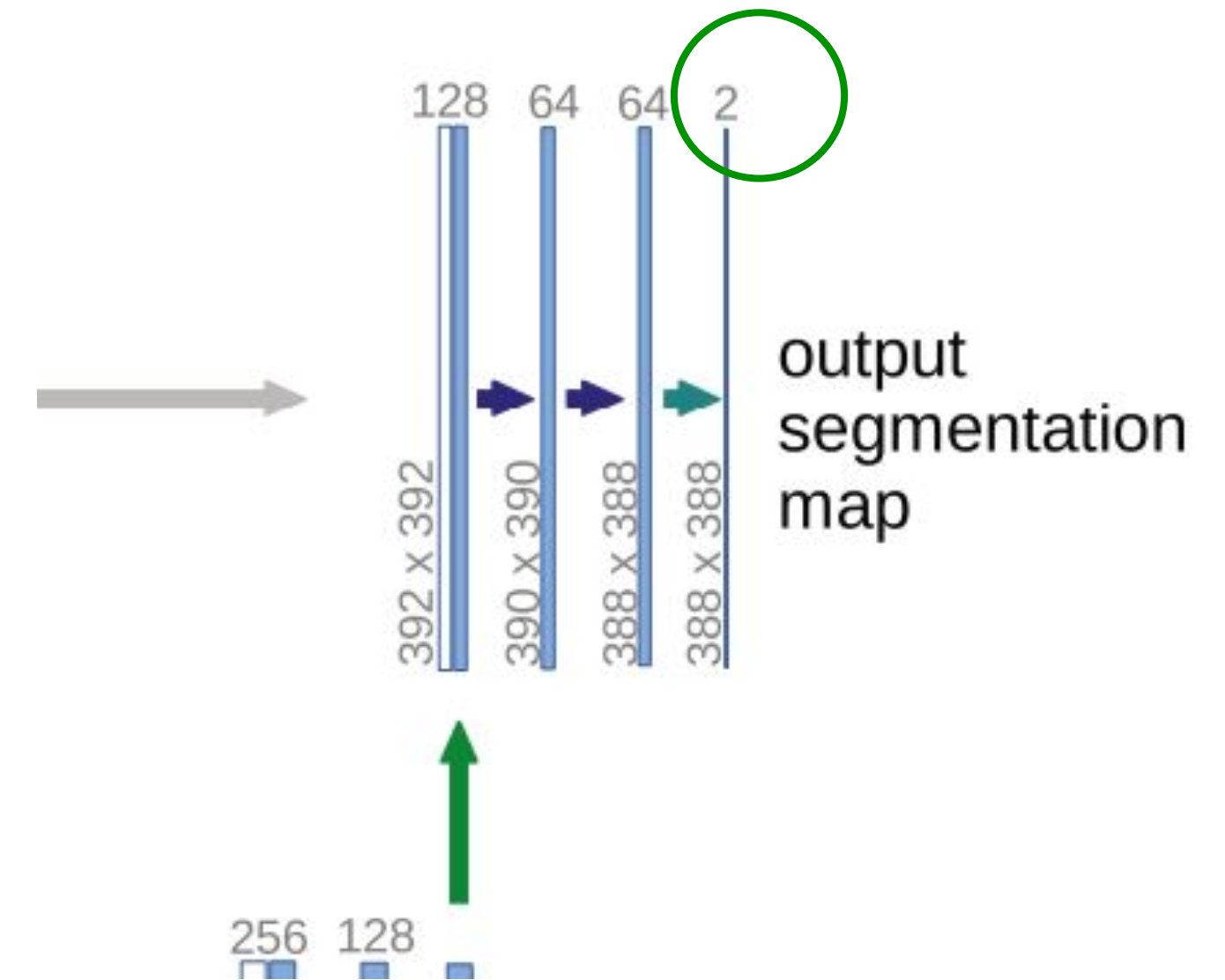


U-Net: Network top

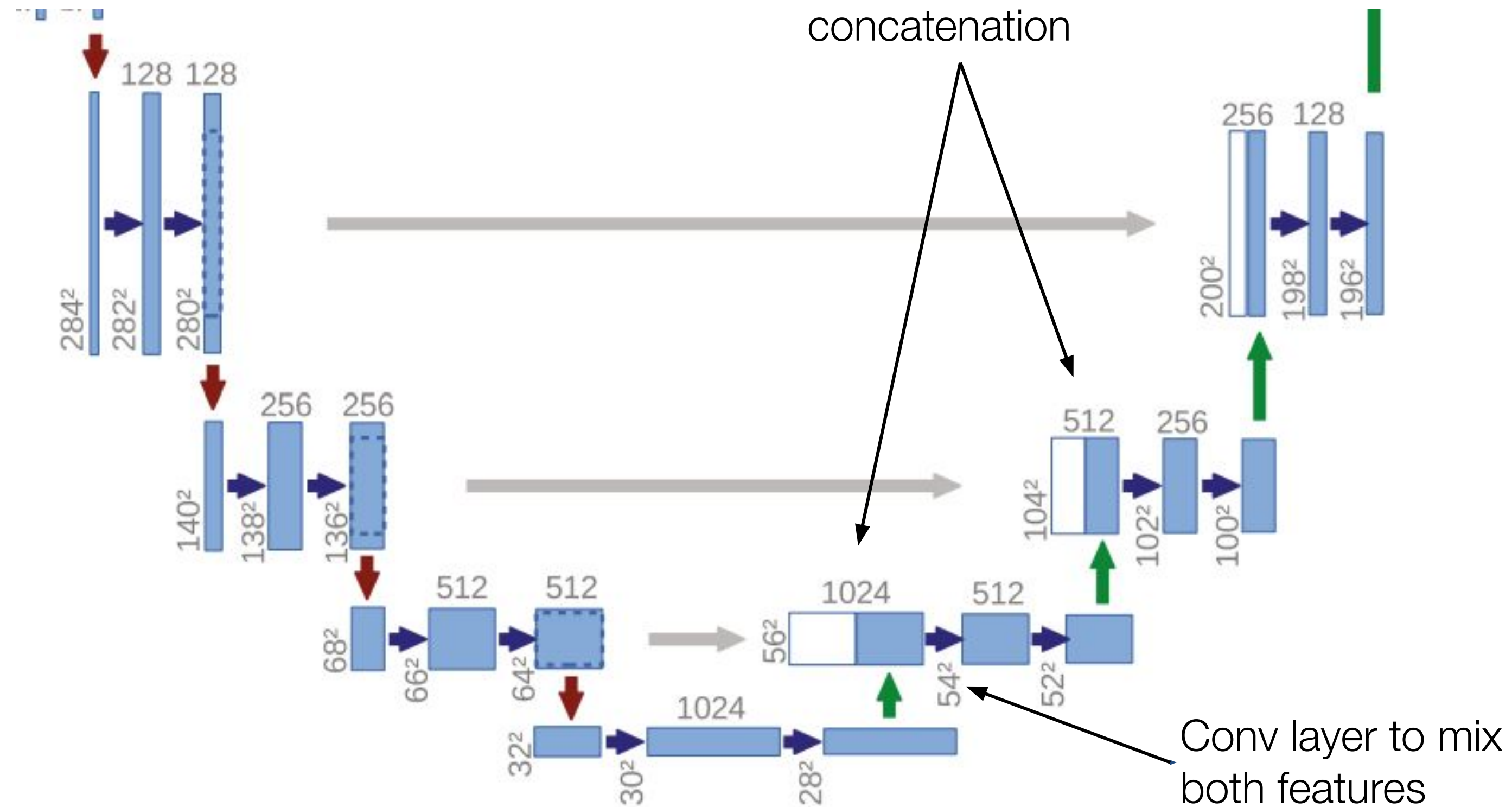
At the final layer, a 1×1 convolution is used to map each 64-channel feature vector into the desired number of output classes.

This operation mixes information across channels while preserving the spatial resolution, so that every pixel location produces its own set of class scores.

After the SoftMax (or Sigmoid for binary segmentation), these scores become per-pixel class probabilities, forming the final segmentation map



U-Net: Skip connections



Specific challenges of segmentation

With segmentation, we are no longer making one prediction per image, but pixel-level predictions instead. This completely changes how training process is monitored and how performance is measured.

- Pixel-wise predictions create extreme class imbalance.
- Accuracy becomes misleading.
- Tiny object get overwhelmed by background (Tversky Loss; Focal Loss);
- There are other aspects to pay attention to (e.g., borders become critical, for example in medical application; e.g., Boundary Loss).

For this reason, proper metrics that capture the overlap and boundary precision must be introduced.

Evaluation metrics

- **Dice coefficient:** metric used to evaluate the overlap between a predicted segmentation mask and a ground truth mask. It ranges from 0 (no overlap) to 1 (perfect overlap).

$$\text{Dice} = \frac{2|\text{True} \cap \text{Predicted}|}{|\text{True}| + |\text{Predicted}|}$$

- **Intersection over Union (IoU):** another metric to measure the overlap between the predicted mask and the ground truth one, comparing the intersection and the union of the predicted and true sets

$$\text{IoU} = \frac{|\text{True} \cap \text{Predicted}|}{|\text{True} \cup \text{Predicted}|}$$

Evaluation metrics

- **Hausdorff Distance:** The Hausdorff distance is a measure of the degree of mismatch between two sets. It is used to compare two shapes. The Hausdorff distance is the maximum distance from a point in one set to the closest point in the other set

$$d_{\Pi}(X, Y) = \max \left(\sup_{x \in X} d(x, Y), \sup_{y \in Y} d(X, y) \right)$$

- **Panoptic Quality:** metric designed to evaluate the performance of panoptic segmentation models. It is computed in two steps: **segment matching** and **quality assessment**. Predicted segments are matched to ground-truth segments based on their Intersection over Union (IoU). Segments with a high IoU are considered True Positives (TP), while unmatched predictions are classified as False Positives (FP) and missing ground-truth segments as False Negatives (FN). The PQ score is then calculated as the product of : **Segmentation Quality (SQ)**, which measures the average IoU of all TP matches, and **Recognition Quality (RQ)**, which is the F1 score for the segments.

$$PQ = \underbrace{\frac{\sum_{(p,g) \in TP} \text{IoU}(p, g)}{|TP|}}_{\text{segmentation quality (SQ)}} \times \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{recognition quality (RQ)}}$$

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